THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains why one factor is changed.
2	**(c)** A is true but R is false.	A is true; R is false because animal cells lack a cell wall.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(d)** greater	12 degrees C versus 7 degrees C.
15	**(d)** 2 mm	2 x 1000 = 2000 micrometres = 2 mm.
16	**(a)** 70	40 + 30 = 70.
17	**(a)** Both A and R are true and R is the correct explanation of A.	Equal thickness is a controlled variable; R is the rule of a fair test.
18	**(a)** Both A and R are true and R is the correct explanation of A.	The compartments he saw were the walls of dead plant cells, which is why they looked like small rooms.
19	**(d)** A is false but R is true.	A is false because health is complete well-being; R is true.
20	**(a)** Both A and R are true and R is the correct explanation of A.	Oersted's observation is explained by the magnetic field of the current.

SECTION — B (Very Short Answer) — Answers 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	T = 34 / 20 = 1.7 s (1); Timing many swings reduces the error of reaction time (1).	1+1
22	Actual = 5 / 500 = 0.01 mm (1); = 10 micrometres (1).	1+1
23	Unvaccinated = 2000 = 25% (1); Fewer susceptible persons means less spread (1).	1+1
24	Both feet are at the same potential (1); No current passes through the bird (1).	1+1
25	Mass remains 3 kg (1); Weight = 30 / 6 = 5 N (1).	1+1
26	Area = 0.005 square metre (1); P = 50/0.005 = 10000 Pa (1).	1+1

SECTION — C (Short Answer) — Answers

6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Only one soil type was compared (1); The conclusion over-generalises (1); Test several soils with controls (1).	1+1+1
28	Onion cells are rectangular with a cell wall; cheek cells are rounded or irregular without a wall (1); Both show a nucleus and cytoplasm; the onion cell has a large vacuole (1); Stains such as safranin or methylene blue make the parts visible (1).	1+1+1
29	Contaminated drinking water (1); Unhygienic food (1); Preventive: boiled water, hygiene, treat water tank (1).	1+1+1
30	Increase the number of turns (1); Increase the number of cells (1); Use a thicker soft iron core (1).	1+1+1
31	Friction = 30 N (1); With 20 N net force = 10 N backwards (1); The box slows down and stops (1).	1+1+1
	OR: Area = 0.01 square m; P = 50/0.01 = 5000 Pa (1); Stacked: weight 100 N (1); P = 10000 Pa (1).	1+1+1
32	-10 to 0: ice warms (1); 0 to 100: melting at 0 then water warms (1); At 100: boiling then steam warms to 110 (1).	1+1+1
	OR: Probably iron, an element (1); Magnetic and reacts with acid giving hydrogen (1); Could also be a mixture containing iron (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) Amount of water in the dough (1). (ii) Time taken to puff (or degree of puffing) (1). (iii) Quantity of flour / thickness of puri / oil temperature (any one) (1). (iv) Hypothesis: puris made with more water puff in less time (or puff better) (1). Repeating trials checks for chance errors and makes the result reliable (1).	1+1+1+2
34	(i) A thin peel lets light pass through so one layer of cells can be seen (1). (ii) Nucleus (1). (iii) Cell wall (1). (iv) Cheek cells have no cell wall and are irregular/rounded in shape, and have no large central vacuole (any two) (2).	1+1+1+2
35	Contaminated food or water at the feast (1); Salmonella bacteria from a carrier or dirty water (1); Hands and flies spread germs (1); Ensure clean water, cooked food, hygiene (1); Treatment with antibiotics on advice; vaccination (1).	1+1+1+1+1
36	Independent: number of turns; dependent: pins lifted (1); Constant: cell, nail, wire (1); Procedure: 10, 20, 30, 40 turns and count pins (1); Expected: pins rise with turns (1); Conclusion (1).	1+1+1+1+1
	OR: Independent: weight on block (1); Dependent: pull on spring balance at constant speed (1); Constant: surface and speed (1); Table of load and friction (1); Graph is a straight line rising (1).	1+1+1+1+1
37	Pressure at 3 m is three times that at 1 m (1); Ears feel more pressure at depth (1); Dam base is thick (1); Pressure acts in all directions (1); Independent of container shape (1).	1+1+1+1+1
	OR: Independent: temperature of water (1); Dependent: time taken for ink to spread (1); Constants: volume, ink drop, beaker (1); Observation: spreads faster in warm water (1); Conclusion: higher temperature gives faster diffusion (1).	1+1+1+1+1
38	Prepare a mixture of iron and sulphur (1); Test with a magnet before heating (1); Heat strongly in a test tube (1); Black, non-magnetic product forms after heating (1); A compound is formed; mixtures retain properties (1).	1+1+1+1+1
	OR: At 70: 150 g solution = 50 g salt + 100 g water (1); At 20 the 100 g water holds 20 g (1); Crystals = 50 - 20 (1); = 30 g (1); Percentage of solution at 20: 20/120 = 16.7 per cent (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 36 g, 1 : 8	4 + 32 = 36; 4:32 = 1:8.
2	**(b)** 39 per cent	64/164 x 100 = 39.
3	**(d)** 140 degrees	Initial i = 55; new i = 70; angle = 140.
4	**(c)** 15 m	3/4 = h/20.
5	**(b)** snake	Energy decreases at each step.
6	**(a)** 0.53	3400/6400.
7	**(b)** 50 degrees C	Subtract zero error of 3.
8	**(a)** stops functioning properly	Nucleus controls the cell.
9	**(d)** colds are viral and often recover by themselves	Colds go away without them.
10	**(c)** the iron rod	Iron concentrates the field.
11	**(d)** moves towards team A	Net force towards A.
12	**(a)** A is likely to be stronger	Sea supplies energy.
13	**(c)** temperature increases the motion of particles	Faster particles.
14	**(d)** homogeneous mixture	Alloy.
15	**(b)** Y	Immiscible liquids.
16	**(d)** 90 degrees to the surface (back along normal)	Incidence along the normal.
17	**(a)** Both A and R are true and R is the correct explanation of A.	R states exactly why friction is a contact force.
18	**(a)** Both A and R are true and R is the correct explanation of A.	The sharp edge has a smaller area, so the pressure is greater.
19	**(a)** Both A and R are true and R is the correct explanation of A.	Sugar particles occupy the inter-particle spaces of water.
20	**(a)** Both A and R are true and R is the correct explanation of A.	R is the definition of an element.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Gas particles move rapidly in all directions (1); They spread uniformly and fill the container (1).	1+1
22	Oxygen = 36 - 4 = 32 g (1); Ratio = 4 : 32 = 1 : 8 (1).	1+1
23	Percentage = 37.5 per cent (1); Water = 62.5 g (1).	1+1
24	Initial i = 60 degrees (1); New i = 70 or 50 degrees (1).	1+1
25	365.25 is slightly more than the actual 365.2422 (1); 1900 is not a leap year but 2000 is (1).	1+1
26	Grass is the producer; deer the primary consumer; tiger the secondary consumer (top carnivore) (1); The tiger level has the least energy (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	NaCl: 4 g undissolved (1); KNO3: 8 g undissolved (1); NaCl is more soluble (1).	1+1+1
28	n = 360/theta - 1 = 7 (1); 360/theta = 8 (1); theta = 45 degrees (1).	1+1+1
29	At noon: almost no shadow (1); At 3 pm: a shadow pointing east (1); Length is larger than at noon (1).	1+1+1
30	Energy decreases at each trophic level (1); A small amount of energy reaches the top level (1); So only a few top carnivores can be supported (1).	1+1+1

31	Dense iron sank to the centre (1); Lighter material floated up (1); Seismic S-waves do not pass through the outer core (1).	1+1+1
	OR: Straight line (1); Passes through the origin (1); Ratio stays constant (1).	1+1+1
32	Digestion: food vacuole (1); Excretion: contractile vacuole (1); Movement: pseudopodia (1).	1+1+1
	OR: Goitre: iodised salt (1); Rickets: milk, sunlight (1); Anaemia: leafy vegetables (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) Malaria (1). (ii) Female Anopheles mosquito (1). (iii) A protozoan (Plasmodium) parasite (1). (iv) Do not let water stagnate (empty coolers, tyres, pots weekly); use mosquito nets/repellents; spray insecticide or oil on ponds (any two) (2).	1+1+1+2
34	(i) So that current flows through the turns and not directly between neighbouring turns/nail (1). (ii) More turns give a stronger electromagnet (8 to 17 pins) (1). (iii) It loses its magnetism (soft iron is a temporary magnet) (1). (iv) Use more cells (more current) / use a larger soft iron core (any two sensible) (2).	1+1+1+2
35	Herbivores increase (1); Overgrazing of grass (1); Soil erosion (1); Water and shelter loss (1); Ecosystem degrades; other species decline (1).	1+1+1+1+1
36	Inner core = 6400 - (35 + 2900 + 2200) (1); = 1265 km (1); Crust 0.5 per cent, mantle 45 per cent (1); Outer core 34 per cent, inner core 20 per cent (1); Mantle is the thickest (1).	1+1+1+1+1
	OR: Curiosity starts the questions (1); Careful observation gives evidence (1); Persistence helps overcome failed trials (1); Repetition ensures reliability (1); Results shared for checking (1).	1+1+1+1+1
37	Cells in a multicellular body divide the work among themselves (1); Nerve cells are long and branched to carry messages (1); Red blood cells carry oxygen (1); Root hair cells have long projections to absorb water (1); Specialisation makes the body work efficiently (1).	1+1+1+1+1
	OR: Pathogen causes disease; host harbours it (1); Vector carries it (1); Symptoms felt, immunity resists (1); Vaccine prevents; antibiotic treats bacteria (1); Epidemic is a sudden spread (1).	1+1+1+1+1
38	Cell about 1.5 V (1); Battery: several cells, higher voltage (1); Mains about 220 V (1); Low voltage cannot drive a harmful current through the body (1); Mains can cause fatal shock (1).	1+1+1+1+1
	OR: Streamlined shape reduces drag (1); Birds and fish (1); Cars and trains (1); Aeroplanes (1); Boats and bullets (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Chloroplasts carry out photosynthesis, which is why plants (which make their own food) have them.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Antibodies prepare the body.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Series connection allows it to sense all current.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Weight equals air resistance.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
13	**(b)** B and C	Only light differs between B and C.
14	**(a)** 0.05 mm	5 / 100 = 0.05 mm.
15	**(d)** 75%	15000/20000 x 100 = 75%.
16	**(c)** glows dimmer	Reversed cell opposes the others.
17	**(a)** Both A and R are true and R is the correct explanation of A.	Very small particles cannot scatter light, so there is no Tyndall effect.
18	**(a)** Both A and R are true and R is the correct explanation of A.	A virtual image is formed by rays that only appear to meet, so it cannot be caught on a screen.
19	**(a)** Both A and R are true and R is the correct explanation of A.	A higher Sun casts a shorter shadow.
20	**(a)** Both A and R are true and R is the correct explanation of A.	Making food from simple substances is what makes them producers.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	0.15 x 1.08 x 10^12 (1); = 1.6 x 10^11 cubic km (1).	1+1
22	Accuracy is closeness to the true value; precision is closeness of repeated readings (1); Readings may be precise but wrong due to zero error (1).	1+1
23	Amoeba has only a flexible membrane (1); Plant cell has a rigid wall (1).	1+1
24	They cannot be cured completely (1); Treatment and lifestyle changes keep them in control (1).	1+1
25	Total resistance increases (1); Less current for each (1).	1+1
26	Friction is lower (1); Less force opposes the sliding (1).	1+1

SECTION — C (Short Answer) — Answers

6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Wire resistance is high (1); Less current reaches the bulb (1); Use a short thick wire (1).	1+1+1
28	Friction depends on the force pressing surfaces (1); Larger area but less pressure per unit area (1); These balance out (1).	1+1+1
29	Pressure gradient is steepest (1); Air rushes toward the centre (1); Rotation increases speed (1).	1+1+1
30	Ice: least energy, close and fixed (1); Water: medium (1); Steam: highest, far apart (1).	1+1+1
31	In diamond all electrons are held in fixed bonds (1); In graphite some electrons are free to move (1); Different arrangements (1).	1+1+1
	OR: Dissolve impure crystals in hot water (1); Filter the hot solution (1); Cool slowly to get pure crystals (1).	1+1+1
32	Close: virtual, erect and enlarged image (1); Far: real and inverted image (1); So the face appears inverted at a distance (1).	1+1+1
	OR: Earth rotates 360 degrees in 24 hours (1); 15 degrees per hour (1); Local noon differs with longitude (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) 60 - 40 = 20 N in the direction of the pull (1). (ii) Moving, since the net force is not zero (1). (iii) Opposite to the direction of motion (1). (iv) 60 - 10 = 50 N (1); rolling friction is much smaller than sliding friction, so less force is wasted (1).	1+1+1+2
34	(i) Pressure = force / area (1). (ii) Area = 0.02 square metre, P = 600 / 0.02 = 30000 Pa (1). (iii) P = 600 / 0.01 = 60000 Pa (1). (iv) A camel's wide feet spread its weight over a large area, so the pressure on the sand is small; a heel has a small area, so the pressure is large (2).	1+1+1+2
35	Oxygen: molecules of two oxygen atoms (1); Water: molecules with H and O atoms bonded (1); Air: different molecules mixed (1); Atoms joined in compounds (1); Mixtures are not bonded (1).	1+1+1+1+1
36	Boil solution in a flask (1); Vapour passes through a condenser (1); Pure water collects (1); Salt remains (1); No dissolved impurities (1).	1+1+1+1+1
	OR: Cornea and lens focus light (1); Image forms on retina (1); Myopia: concave lens (1); Hypermetropia: convex lens (1); Cataract and other causes (1).	1+1+1+1+1
37	Latitude = 30 degrees N (1); Equator: on the horizon (1); North Pole: overhead (1); Height increases as he travels north (1); Pole Star altitude equals latitude (1).	1+1+1+1+1
	OR: Global warming (1); Ice melts and sea level rises (1); Ocean acidification (1); Extreme weather (1); Individual: save energy and plant trees (1).	1+1+1+1+1
38	Atmosphere: warming (1); Oceans: more acidic, coral damage (1); Plants: faster growth up to a point (1); But heat and drought reduce crops (1); Net effect is harmful (1).	1+1+1+1+1
	OR: Height increases up to 60 mL (1); Falls at 80 mL (1); Too much water may reduce air in the soil (1); Next test: values between 40 and 80 mL (1); Repeat with several plants (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.